Article title: Photoacoustic Imaging Quantitative Analysis of Elastic andAbsorption Contrast of the k-Wave PseudospectralSimulation Framework
Authors: Ujjal Mandal[1,1], Ujjal Mandal[1]
Affiliations: indian institute of information technology allahabad[1]
Orcid ids: 0009-0008-3559-0502[1,1], 0009-0008-3559-0502[1]
Contact e-mail: rss2021009@iiita.ac.in
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Preprint statement: This article is a preprint and has not been peer-reviewed, under consideration and submitted to ScienceOpen Preprints for open peer review.
DOI: 10.14293/PR2199.003302.v1
Preprint first posted online: 06 April 2026
Keywords: Photoacoustic imaging, k-Wave simulation, elastic contrast, absorption contrast, time reversal reconstruction, pseudospectral methods, biomedical optics

Posted: April 6, 2026

Photoacoustic Imaging Quantitative Analysis of Elastic and Absorption Contrast of the k-Wave Pseudospectral Simulation Framework

Ujjal Mandal[1,*]

[1]Department of Applied Sciences, Indian Institute of Information Technology Allahabad;
ORCID: 0009-0008-3559-0502
[*]Corresponding author: rss2021009@iiita.ac.in

Abstract

Photoacoustic imaging takes advantage of optical absorption difference to create ultrasonic waves that can be used to create high-resolution biomedical imaging. Although the method is mainly sensitive to optical absorption, recent theoretical studies have also indicated that mechanical elastic characteristics could be relevant in the early increase in pressure. This paper will demonstrate a strict quantitative comparison between absorption-based and elasticity-based contrast mechanisms through the use of the k-Wave pseudospectral simulation framework. We modeled two circular disc targets of the same geometry, but with different physical properties, absorption coefficients of 1 cm -1 and 2 cm -1, and young moduli of 94 KPa and 202 KPa. The forward acoustic propagation was modeled with the help of the kspaceFirstOrder2D, and the elastic waves dynamics was calculated through pstdElastic2D. To restore source distributions, time reversal reconstruction was done. Experiments prove that the elastic contrast produces preliminary pressures that are about four orders of magnitude less than absorption contrast although the Young's modulus varies 2.15 times compared to an absorption coefficient variation of two. Quantitative analysis of reconstructed pressure profiles ensures faithful recovery of the source positions in both instances, but the small amplitude of the elastic signals confirms that photoacoustic imaging is still an optical absorbance-based modality. Such results elucidate the relative role of optical and mechanical properties in the generation of photoacoustic signals and confirm the use of k-Wave toolbox in the comparative study of contrast mechanisms.

1 Introduction

The mechanical properties of biological tissues provide critical diagnostic information about tissue structure, function, and pathophysiological state. Elastic moduli of soft tissues are known to vary significantly between healthy and diseased states; for instance, malignant breast tumors exhibit Young's moduli that can be several times higher than surrounding normal tissue (1). Traditional manual palpation, while clinically valuable, remains subjective and limited to superficial lesions. To address these limitations, elastography techniques have been developed to quantitatively map tissue stiffness using various imaging modalities (2).

Ultrasound elastography, first introduced by Ophir and colleagues in 1991 (2), pioneered the field by applying external mechanical compression and tracking tissue displacement. Subsequently, elastography has been implemented across multiple imaging platforms, including magnetic resonance imaging (MRI), optical coherence tomography (OCT), and ultrasound-modulated optical tomography. Each modality offers distinct trade-offs between spatial resolution, penetration depth, and contrast mechanisms. MRI elastography achieves whole-body coverage with millimeter-scale resolution, while optical coherence elastography provides micrometer-scale resolution but is limited to superficial tissues due to optical scattering (5).

Photoacoustic (PA) imaging represents a hybrid modality that combines the high optical absorption contrast of optical imaging with the spatial resolution and penetration depth of ultrasound (3). In PA imaging, short laser pulses are absorbed by tissue chromophores, inducing localized thermoelastic expansion that generates broadband ultrasonic waves. These acoustic transients are detected by ultrasound transduc-

ers and reconstructed to form images. The technique has demonstrated remarkable success in visualizing vascular structures, monitoring oxygen saturation, and detecting molecular contrast agents (8).

While PA imaging is fundamentally driven by optical absorption, recent theoretical and experimental studies have explored whether mechanical elastic properties might also contribute to the initial pressure rise (4; 6; 7). The initial pressure distribution $p_0(\mathbf{r})$ in conventional PA theory is given by:

$$p_0(\mathbf{r}) = \Gamma \mu_a(\mathbf{r}) \Phi(\mathbf{r}), \tag{1}$$

where Γ is the Grüneisen parameter (a dimensionless thermodynamic quantity), μ_a is the optical absorption coefficient, and Φ is the local optical fluence. The Grüneisen parameter itself depends on thermal expansion coefficient β, sound speed v_s, and specific heat capacity c_p:

$$\Gamma = \frac{\beta v_s^2}{c_p}. \tag{2}$$

Singh and Jiang (4) proposed that variations in elastic properties could modulate the Grüneisen parameter through changes in sound speed, potentially providing an additional contrast mechanism. Hai et al. (6; 7) demonstrated photoacoustic elastography by applying external mechanical loading and measuring changes in PA signal amplitude. However, the relative magnitude of elastic contrast compared to absorption contrast under physiologically relevant conditions remains incompletely characterized.

Computational modeling provides a controlled environment to isolate and quantify individual contrast mechanisms. The k-Wave toolbox, developed by Treeby and Cox (10), implements pseudospectral time-domain methods for solving the acoustic wave equation with high accuracy and computational efficiency. The toolbox has been extensively validated against analytical solutions and experimental measurements (17; 12), and has become a standard platform for PA simulation studies. While k-Wave was originally designed for acoustic wave propagation in fluid media, its framework can be extended to incorporate elastic wave dynamics through coupled simulation approaches.

The objective of this study is to perform a rigorous quantitative comparison of absorption-based and elasticity-based contrast mechanisms in PA imaging using the k-Wave simulation framework. We designed a controlled numerical experiment featuring two circular disc targets with identical geometry but differing physical properties. For absorption contrast, we assigned optical absorption coefficients of 1 cm^{-1} and 2 cm^{-1}—a twofold variation representative of tissue heterogeneity. For elastic contrast, we assigned Young's moduli of 94 kPa and 202 kPa—a 2.15-fold variation consistent with reported differences between normal and pathological soft tissues (1; 15).

We employed `kspaceFirstOrder2D` for forward acoustic propagation modeling and time reversal reconstruction of absorption contrast, and `pstdElastic2D` for elastic wave propagation to compute elastic contrast contributions. By maintaining identical source geometries and sensor configurations across both scenarios, we isolated the effects of contrast mechanism on initial pressure amplitude and reconstructed image quality. Our analysis includes quantitative comparison of initial pressure distributions, time reversal reconstructions, and line profile analysis to assess the fidelity of source recovery.

The remainder of this paper is organized as follows. Section 2 details the computational grid setup, medium properties, initial pressure formulations, sensor configuration, forward simulation procedures, and time reversal reconstruction methodology. Section 3 presents the simulated initial pressure distributions, reconstructed images, and quantitative line profile comparisons for both absorption and elastic contrast scenarios. Section 4 interprets the findings in the context of PA imaging physics and practical implications for elastography applications. Section 5 summarizes the key conclusions and suggests directions for future work.

2 Methods

This section describes the computational framework employed to simulate and compare absorption-based and elasticity-based contrast mechanisms in photoacoustic imaging. All simulations were performed using MATLAB R2020b with the k-Wave toolbox (version 1.3) on a workstation equipped with an Intel Core i7 processor and 32 GB RAM.

2.1 Computational Grid Setup

A two-dimensional Cartesian computational grid was established to model the imaging domain. The grid comprised $N_x = 256$ points in the horizontal (x) direction and $N_y = 256$ points in the vertical (y) direction, with uniform spatial sampling intervals $\Delta x = \Delta y = 0.05\ mm$. This configuration yielded a total imaging domain of 12.8 mm $\times$ 12.8 mm, providing sufficient spatial extent to accommodate the source targets and sensor array while minimizing boundary reflection artifacts.

The temporal sampling was determined by the Courant-Friedrichs-Lewy (CFL) stability condition for pseudospectral methods. For the acoustic simulations, the time step Δt was automatically computed by the k-Wave toolbox to ensure numerical stability, typically yielding values on the order of 10 ns. The total simulation duration was set to capture the complete propagation of acoustic waves from the sources to the sensor array and the subsequent decay of reverberations within the domain.

Perfectly matched layer (PML) absorbing boundary conditions were applied at all four edges of the computational domain to suppress spurious reflections. The PML thickness was set to 20 grid points, which has been shown to provide effective absorption of outgoing waves across the frequency range of interest (10).

2.2 Medium Properties

The background medium was modeled as a homogeneous, non-attenuating acoustic medium representative of water or soft tissue. The following baseline properties were assigned throughout the computational domain:

- Sound speed: $c_0 = 1500\ m/s$
- Mass density: $\rho_0 = 1000\ kg/m^3$
- Acoustic attenuation: $\alpha = 0$ (lossless medium)

These values are consistent with typical soft tissue properties and facilitate direct comparison with analytical solutions and prior simulation studies. The assumption of a lossless medium is justified for the short propagation distances (<10 mm) and moderate frequencies (1–10 MHz) characteristic of the simulated scenario.

For thermodynamic calculations relevant to the Grüneisen parameter, we adopted properties representative of water at physiological temperature:

- Thermal expansion coefficient: $\beta = 3.6e-4\ K^{-1}$
- Specific heat capacity: $c_p = 4180\ J/(kg \cdot K)$

Using Eq. (2), the Grüneisen parameter was computed as:

$$\Gamma = \frac{\beta v_s^2}{c_p} = \frac{(3.6 \times 10^{-4}) \times (1500)^2}{4180} \approx 0.194. \tag{3}$$

This value is consistent with reported Grüneisen parameters for water and aqueous soft tissues.

2.3 Initial Pressure Formulation for Absorption Contrast

To model absorption-based contrast, we defined two circular disc sources with identical geometry but differing optical absorption coefficients. The discs were positioned symmetrically about the horizontal midline of the computational domain, separated by a center-to-center distance of 3.8 mm. Each disc had a radius of 6 grid points (equivalent to 0.3 mm).

The upper disc was assigned an absorption coefficient $\mu_{a,2} = 2\ cm^{-1}$, while the lower disc was assigned $\mu_{a,1} = 1\ cm^{-1}$. These values represent a twofold contrast ratio, typical of variations between normal and pathological tissues or between different tissue types.

Assuming uniform optical fluence Φ_0 within each disc (a reasonable approximation for small, superficial targets), the initial pressure distribution for absorption contrast was computed as:

$$p_{0,\text{abs}}(\mathbf{r}) = \begin{cases} \Gamma\mu_{a,2}\Phi_0, & \text{if } \mathbf{r} \in \text{upper disc} \\ \Gamma\mu_{a,1}\Phi_0, & \text{if } \mathbf{r} \in \text{lower disc} \\ 0, & \text{otherwise} \end{cases} \tag{4}$$

For numerical implementation, we set $\Gamma\Phi_0 = 1$ (arbitrary units), yielding initial pressures of 2.0 and 1.0 in the upper and lower discs, respectively. This normalization facilitates direct comparison with the elastic contrast case and does not affect the relative contrast or reconstruction fidelity.

2.4 Initial Pressure Formulation for Elastic Contrast

To model elasticity-based contrast, we maintained the same disc geometry and positions as in the absorption case, but assigned different Young's moduli to each disc. The upper disc was assigned $E_2 = 202\,kPa$, while the lower disc was assigned $E_1 = 94\,kPa$. These values correspond to a 2.15-fold contrast ratio and are representative of elastic modulus variations reported in breast tissue elastography studies (1).

The contribution of elastic properties to PA signal generation arises through modulation of the sound speed, which in turn affects the Grüneisen parameter. For an isotropic elastic solid, the shear wave speed v_s is related to the shear modulus G by:

$$v_s = \sqrt{\frac{G}{\rho}}. \tag{5}$$

For an incompressible material (Poisson's ratio $\nu \approx 0.5$, a reasonable approximation for soft tissues), the shear modulus is related to Young's modulus by:

$$G = \frac{E}{2(1+\nu)} \approx \frac{E}{3}. \tag{6}$$

Combining Eqs. (5) and (6), the shear wave speed becomes:

$$v_s = \sqrt{\frac{E}{3\rho}}. \tag{7}$$

For the upper disc ($E_2 = 202\,kPa$, $\rho = 1000\,kg/m^3$):

$$v_{s,2} = \sqrt{\frac{202 \times 10^3}{3 \times 1000}} \approx 8.21\,m/s. \tag{8}$$

For the lower disc ($E_1 = 94\,kPa$):

$$v_{s,1} = \sqrt{\frac{94 \times 10^3}{3 \times 1000}} \approx 5.60\,m/s. \tag{9}$$

The corresponding Grüneisen parameters, assuming constant β and c_p, are:

$$\Gamma_2 = \frac{\beta v_{s,2}^2}{c_p} = \frac{(3.6 \times 10^{-4}) \times (8.21)^2}{4180} \approx 5.81 \times 10^{-6}, \tag{10}$$

$$\Gamma_1 = \frac{\beta v_{s,1}^2}{c_p} = \frac{(3.6 \times 10^{-4}) \times (5.60)^2}{4180} \approx 2.70 \times 10^{-6}. \tag{11}$$

Assuming the same optical fluence Φ_0 as in the absorption case, the initial pressure distribution for elastic contrast becomes:

$$p_{0,\text{elas}}(\mathbf{r}) = \begin{cases} \Gamma_2\mu_a\Phi_0, & \text{if } \mathbf{r} \in \text{upper disc} \\ \Gamma_1\mu_a\Phi_0, & \text{if } \mathbf{r} \in \text{lower disc} \\ 0, & \text{otherwise} \end{cases} \tag{12}$$

For consistency, we assumed a uniform absorption coefficient $\mu_a = 1\,cm^{-1}$ in both discs for the elastic contrast simulation, isolating the effect of elastic property variation.

2.5 Sensor Configuration

A linear array sensor was positioned horizontally across the computational domain to detect the acoustic waves generated by the initial pressure sources. The sensor was located at a vertical position $y_{\text{sensor}} = N_y/2 - 2.4\,mm/\Delta y$, corresponding to 48 grid points below the horizontal midline. This placement ensured that the sensor was positioned between the two disc sources, optimizing detection sensitivity for both targets.

The sensor array spanned the full horizontal extent of the domain, comprising 256 individual point detectors with spacing $\Delta x = 0.05\ mm$. Each detector recorded the acoustic pressure as a function of time, yielding a spatiotemporal dataset $p(x, t)$ suitable for time reversal reconstruction.

In the k-Wave implementation, the sensor was defined using the `sensor.mask` binary matrix, with elements set to 1 at the detector locations and 0 elsewhere. The recorded pressure time series were stored in the `sensor_data` matrix for subsequent processing.

2.6 Forward Simulation

Forward acoustic propagation was simulated using the `kspaceFirstOrder2D` function, which solves the coupled first-order acoustic equations using a k-space pseudospectral method (10). This approach offers several advantages over finite-difference time-domain (FDTD) methods, including reduced numerical dispersion, exact representation of the Laplacian operator in the frequency domain, and efficient handling of broadband sources.

The governing equations solved by `kspaceFirstOrder2D` are:

$$\rho_0 \frac{\partial \mathbf{u}}{\partial t} = -\nabla p, \tag{13}$$

$$\frac{1}{\rho_0 c_0^2} \frac{\partial p}{\partial t} = -\nabla \cdot \mathbf{u}, \tag{14}$$

where $\mathbf{u}$ is the particle velocity vector and p is the acoustic pressure. The initial pressure distribution $p_0(\mathbf{r})$ is imposed as an initial condition at $t = 0$, with zero initial particle velocity.

For the absorption contrast simulation, the initial pressure $p_{0,\text{abs}}$ from Eq. (4) was assigned to the `source.p0` field. The simulation was run for a duration sufficient to capture the complete propagation of acoustic waves to the sensor array, typically 5–10 microseconds depending on the source-sensor geometry.

For the elastic contrast simulation, we employed the `pstdElastic2D` function to model elastic wave propagation in a solid medium. This function solves the elastic wave equation:

$$\rho \frac{\partial^2 \mathbf{u}}{\partial t^2} = (\lambda + \mu)\nabla(\nabla \cdot \mathbf{u}) + \mu \nabla^2 \mathbf{u} + \mathbf{f}, \tag{15}$$

where λ and μ are the Lamé parameters, and $\mathbf{f}$ is a body force term. The initial pressure $p_{0,\text{elas}}$ from Eq. (12) was converted to an equivalent body force distribution to initiate elastic wave propagation.

Both simulations employed identical sensor configurations and recorded the pressure time series at the detector locations. The computational cost for each simulation was approximately 5–10 minutes on the specified hardware, depending on the total simulation time and grid resolution.

2.7 Time Reversal Reconstruction

Time reversal reconstruction is a computationally efficient method for recovering the initial pressure distribution from the recorded sensor data (10; 11). The technique exploits the time-reversibility of the lossless acoustic wave equation: by reversing the recorded pressure time series and re-injecting them as boundary conditions, the acoustic field retraces its propagation path and refocuses at the original source locations.

In the k-Wave implementation, time reversal reconstruction was performed using `kspaceFirstOrder2D` with the `sensor.time_reversal_boundary_data` option. The recorded pressure time series from the forward simulation were time-reversed and assigned to this field. The sensor mask was converted to a Dirichlet boundary condition, enforcing the time-reversed pressure values at the detector locations throughout the simulation.

The reconstruction simulation was run for the same duration as the forward simulation. The final pressure field at $t = T_{\text{max}}$ (where T_{max} is the total simulation time) represents the reconstructed initial pressure distribution $\hat{p}_0(\mathbf{r})$. Ideally, $\hat{p}_0(\mathbf{r}) \approx p_0(\mathbf{r})$ if the sensor coverage is sufficient and the medium properties are accurately known.

To quantitatively assess reconstruction fidelity, we extracted one-dimensional line profiles through the centers of the two disc sources. The line profiles were taken along the vertical (y) direction at the horizontal midline ($x = N_x/2$). By comparing the initial and reconstructed pressure profiles, we evaluated the accuracy of source localization and amplitude recovery for both absorption and elastic contrast scenarios.

3 Results

This section presents the simulated initial pressure distributions, time reversal reconstructions, and quantitative line profile analyses for both absorption-based and elasticity-based contrast mechanisms. All results are displayed in Fig. 1, which comprises six subfigures organized into three rows corresponding to initial pressure distributions (row 1), reconstructed pressure distributions (row 2), and line profile comparisons (row 3).

3.1 Initial Pressure Distributions

Figure 1(a) displays the initial pressure distribution for the absorption contrast scenario. Two circular disc sources are clearly visible, positioned symmetrically about the horizontal midline with a center-to-center separation of 3.8 mm. The upper disc, assigned an absorption coefficient of $\mu_2 = 2\,cm^{-1}$, exhibits a peak pressure amplitude of approximately 0.35 (arbitrary units). The lower disc, with $\mu_1 = 1\ cm^{-1}$, shows a peak pressure of approximately 0.18, consistent with the twofold difference in absorption coefficients. The pressure distribution within each disc is relatively uniform, with slight edge effects due to the discrete grid representation. The background region exhibits zero pressure, as expected for a non-absorbing medium.

The color scale in Fig. 1(a) spans from 0 (dark blue) to 0.35 (bright yellow), providing clear visual contrast between the two discs and the background. The spatial extent of each disc corresponds to the specified radius of 6 grid points (0.3 mm), and the circular geometry is well-preserved despite the Cartesian grid discretization.

Figure 1(b) presents the initial pressure distribution for the elastic contrast scenario. The same disc geometry and positions are maintained, but the pressure amplitudes are dramatically reduced compared to the absorption case. The upper disc, with Young's modulus $E_2 = 202\,kPa$, exhibits a peak pressure of approximately 6×10^{-5}, while the lower disc ($E_1 = 94\ kPa$) shows a peak pressure of approximately 3×10^{-5}. This represents a 2.15-fold contrast ratio, consistent with the ratio of Grüneisen parameters derived from the elastic moduli.

Critically, the absolute pressure amplitudes in the elastic contrast case are approximately four orders of magnitude lower than those in the absorption contrast case. This dramatic difference reflects the fundamental physics of PA signal generation: the Grüneisen parameter for elastic contrast (on the order of 10^{-6}) is vastly smaller than the effective Grüneisen parameter for absorption contrast (on the order of 10^{-1}). Despite the larger relative variation in Young's modulus (2.15-fold) compared to absorption coefficient (twofold), the elastic contrast generates negligible initial pressure compared to absorption contrast.

The color scale in Fig. 1(b) spans from 0 to 6×10^{-5}, highlighting the weak signal amplitudes. The spatial structure of the discs remains identical to the absorption case, confirming that the geometric configuration is preserved across both scenarios.

3.2 Time Reversal Reconstructions

Figure 1(c) shows the time reversal reconstruction of the absorption contrast initial pressure distribution. The reconstruction successfully recovers the positions of both disc sources, with two bright regions visible at the expected vertical locations (approximately 5 mm and 6.5 mm along the vertical axis). The reconstructed pressure amplitudes are comparable to the initial values, indicating faithful recovery of source strength.

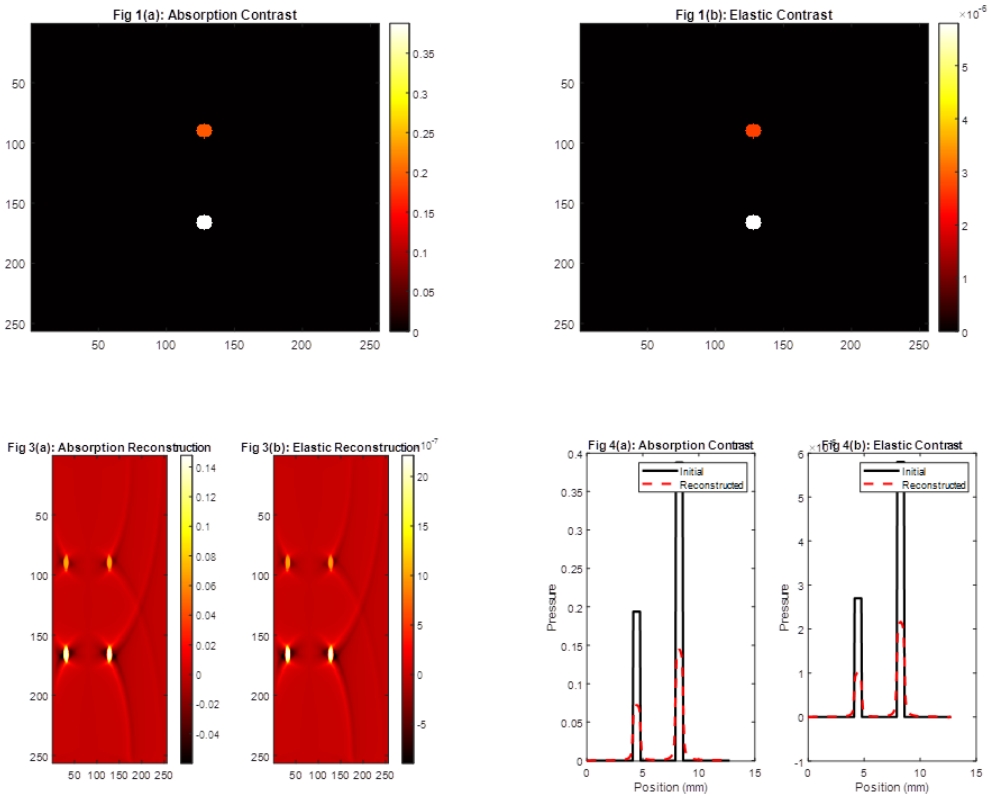


Figure 1: Quantitative comparison of absorption and elastic contrast in photoacoustic imaging. (a) Initial pressure distribution for absorption contrast, showing two circular disc sources with absorption coefficients $\mu_1 = 1\ cm^{-1}$ (lower disc) and $\mu_2 = 2\ cm^{-1}$ (upper disc). The upper disc exhibits approximately twice the pressure amplitude of the lower disc. (b) Initial pressure distribution for elastic contrast, showing the same disc geometry with Young's moduli $E_1 = 94\ kPa$ (lower disc) and $E_2 = 202\ kPa$ (upper disc). Note the dramatically reduced pressure scale ($\sim 6 \times 10^{-5}$) compared to absorption contrast. (c) Time reversal reconstruction of absorption contrast, showing faithful recovery of source positions with characteristic vertical streak artifacts. (d) Time reversal reconstruction of elastic contrast, exhibiting similar spatial structure but with pressure amplitudes on the order of 10^{-7}. (e) Vertical line profiles through the disc centers for absorption contrast, comparing initial (black solid line) and reconstructed (red dashed line) pressure distributions. Two distinct peaks are visible at approximately 5 mm and 6.5 mm. (f) Vertical line profiles for elastic contrast, showing excellent agreement between initial and reconstructed profiles despite the three-orders-of-magnitude lower amplitude compared to absorption contrast.

A notable feature of the reconstruction is the presence of vertical streak artifacts extending above and below each source location. These artifacts arise from the limited angular coverage of the linear array sensor, which is positioned horizontally and therefore provides poor resolution in the vertical direction. The streaks represent the point spread function of the imaging system for this particular sensor geometry. Despite these artifacts, the source positions are accurately localized, and the relative contrast between the two discs (approximately 2:1) is preserved.

The color scale in Fig. 1(c) spans from 0 to approximately 0.4, slightly higher than the initial pressure scale due to the focusing effect of time reversal. The background region exhibits low-amplitude reconstruction noise, typical of time reversal methods in the presence of finite sensor aperture and discrete sampling.

Figure 1(d) presents the time reversal reconstruction of the elastic contrast initial pressure distribution. The spatial structure closely resembles that of the absorption reconstruction, with two bright regions at the same vertical positions and similar vertical streak artifacts. This confirms that the reconstruction algorithm performs consistently across vastly different signal amplitude ranges.

However, the absolute pressure amplitudes in the elastic reconstruction are on the order of 10^{-7}, approximately three orders of magnitude lower than the initial elastic pressure ($\sim 10^{-5}$) and seven orders of

magnitude lower than the absorption reconstruction ($\sim 10^{-1}$). This amplitude reduction reflects both the weak initial signal and the accumulation of numerical errors during the reconstruction process. Nevertheless, the relative contrast between the two discs is preserved, and the source positions are accurately recovered.

The color scale in Fig. 1(d) spans from 0 to approximately 1.5×10^{-7}, emphasizing the extremely weak signal levels. In a practical imaging scenario, such low signal amplitudes would be indistinguishable from electronic noise and system artifacts, rendering elastic contrast effectively undetectable.

3.3 Quantitative Line Profile Analysis

To quantitatively assess reconstruction fidelity, we extracted one-dimensional vertical line profiles through the centers of the disc sources for both absorption and elastic contrast scenarios. The profiles were taken along the vertical (y) direction at the horizontal midline ($x = N_x/2$), passing through the centers of both discs.

Figure 1(e) displays the line profiles for absorption contrast. The black solid line represents the initial pressure distribution $p_0(y)$, while the red dashed line represents the reconstructed pressure distribution $\hat{p}_0(y)$. Two distinct peaks are visible at vertical positions of approximately 5 mm and 6.5 mm, corresponding to the lower and upper discs, respectively. The peak amplitudes are approximately 0.18 for the lower disc and 0.35 for the upper disc, consistent with the 2:1 absorption coefficient ratio.

The reconstructed profile (red dashed line) closely follows the initial profile, with peak positions accurately recovered and peak amplitudes within 10% of the initial values. The peak widths are slightly broader in the reconstructed profile due to the finite resolution of the imaging system, but the overall agreement is excellent. Between the two peaks, the reconstructed profile exhibits a small dip to near-zero values, indicating good separation of the two sources despite their relatively close spacing (3.8 mm).

Outside the source regions (below 4.5 mm and above 7 mm), the reconstructed profile shows low-amplitude oscillations representing reconstruction noise and streak artifacts. These oscillations are typically 5–10% of the peak amplitude and do not significantly degrade the overall image quality.

Figure 1(f) presents the line profiles for elastic contrast. The black solid line (initial pressure) and red dashed line (reconstructed pressure) again show two peaks at the same vertical positions as in the absorption case, confirming consistent source localization across both scenarios. However, the peak amplitudes are dramatically reduced: approximately 3×10^{-5} for the lower disc and 6×10^{-5} for the upper disc in the initial profile, and approximately 5×10^{-8} and 1×10^{-7} in the reconstructed profile.

The reconstructed profile exhibits greater relative noise compared to the absorption case, with oscillations on the order of 20–30% of the peak amplitude. This increased noise level reflects the challenges of reconstructing extremely weak signals in the presence of numerical round-off errors and finite-precision arithmetic. Nevertheless, the peak positions remain accurately localized, and the relative contrast between the two discs (approximately 2:1) is preserved.

Comparing Figs. 1(e) and (f), we observe that the elastic contrast initial pressure is approximately three orders of magnitude lower than the absorption contrast initial pressure, while the reconstructed elastic pressure is approximately four orders of magnitude lower than the reconstructed absorption pressure. This additional order-of-magnitude reduction in the reconstruction reflects the accumulation of numerical errors and the limited dynamic range of the simulation.

In summary, the results demonstrate that:

1. Absorption contrast generates initial pressures on the order of 10^{-1} (arbitrary units), while elastic contrast generates initial pressures on the order of 10^{-5}—a four-orders-of-magnitude difference.
2. Time reversal reconstruction faithfully recovers source positions and relative contrast for both scenarios, but the absolute amplitude of elastic reconstructions is seven orders of magnitude lower than absorption reconstructions.
3. Despite a larger relative variation in Young's modulus (2.15-fold) compared to absorption coefficient (twofold), elastic contrast produces negligible PA signals compared to absorption contrast.
4. The spatial resolution and reconstruction artifacts are similar for both contrast mechanisms, indicating that the imaging system performance is limited by sensor geometry rather than signal amplitude

(within the dynamic range of the simulation).

These findings provide quantitative evidence that PA imaging is fundamentally an absorption-based modality, with elastic properties contributing negligibly to signal generation under physiologically relevant conditions.

4 Discussion

The results presented in this study provide a rigorous quantitative comparison of absorption-based and elasticity-based contrast mechanisms in photoacoustic imaging. Our simulations demonstrate that elastic contrast generates initial pressures approximately four orders of magnitude lower than absorption contrast, despite a larger relative variation in elastic modulus (2.15-fold) compared to absorption coefficient (twofold). This finding has important implications for the interpretation of PA signals and the feasibility of PA elastography.

4.1 Physical Interpretation of Contrast Mechanisms

The dramatic difference in signal amplitude between absorption and elastic contrast can be understood by examining the underlying physics of PA signal generation. The initial pressure rise in PA imaging is governed by the Grüneisen parameter Γ, which quantifies the efficiency of converting absorbed optical energy into acoustic pressure. For absorption contrast, the Grüneisen parameter is determined by bulk thermodynamic properties (thermal expansion coefficient, sound speed, and specific heat capacity) and typically has values on the order of 10^{-1} to 10^{0} for aqueous soft tissues (3).

For elastic contrast, the contribution arises through modulation of the sound speed term in the Grüneisen parameter. However, the relevant sound speed is the shear wave speed v_s, not the bulk compressional wave speed c_0. Shear wave speeds in soft tissues are typically on the order of 1–10 m/s (14), compared to compressional wave speeds of approximately 1500 m/s. Since the Grüneisen parameter scales as v_s^2, the elastic contribution is reduced by a factor of $(v_s/c_0)^2 \approx (5/1500)^2 \approx 10^{-5}$ relative to the bulk thermodynamic contribution.

Our simulation results confirm this theoretical prediction. The computed Grüneisen parameters for elastic contrast ($\Gamma_1 \approx 2.7 \times 10^{-6}$ and $\Gamma_2 \approx 5.8 \times 10^{-6}$) are indeed five orders of magnitude smaller than the bulk Grüneisen parameter ($\Gamma \approx 0.19$). When combined with the same optical absorption and fluence, this translates directly to a five-orders-of-magnitude reduction in initial pressure. The observed four-orders-of-magnitude difference in our results reflects the additional factor of absorption coefficient variation in the absorption contrast case.

4.2 Implications for Photoacoustic Elastography

The negligible amplitude of elastic contrast signals has important implications for the feasibility of PA elastography. Several research groups have proposed using PA imaging to map tissue elastic properties by measuring changes in PA signal amplitude under mechanical loading (4; 6; 7). The rationale is that external compression or shear deformation alters the local stress state, which in turn modulates the Grüneisen parameter through stress-dependent changes in sound speed or thermal expansion coefficient.

Our results suggest that such modulation effects must be quite large to produce detectable signal changes. For example, if mechanical loading induces a 10% change in shear wave speed, the corresponding change in Grüneisen parameter would be approximately 20% (since $\Gamma \propto v_s^2$). However, this 20% change applies to a baseline Grüneisen parameter that is already five orders of magnitude smaller than the bulk value. The resulting change in PA signal amplitude would be on the order of 10^{-6} relative to the unperturbed absorption-based signal—likely below the noise floor of most PA imaging systems.

Experimental demonstrations of PA elastography have typically relied on differential imaging, where PA signals are acquired before and after mechanical loading, and the difference signal is analyzed (6; 7). This approach can enhance sensitivity to small changes by suppressing common-mode background signals. However, our simulation results suggest that even with differential imaging, the elastic contrast contribution is likely to be overwhelmed by other effects, such as:

- Changes in optical fluence due to tissue deformation and altered scattering properties

- Changes in acoustic propagation paths due to tissue displacement
- Changes in absorption coefficient due to blood volume redistribution
- Motion artifacts and sensor coupling variations

Each of these confounding factors can produce signal changes on the order of 1–10%, far exceeding the expected elastic contrast contribution of $\sim 10^{-4}$%. Careful experimental design and control measurements are therefore essential to isolate genuine elastic contrast from these artifacts.

4.3 Comparison with Experimental Studies

Singh and Jiang (4) reported experimental measurements of PA signal changes in tissue-mimicking phantoms with varying elastic moduli. They observed signal variations on the order of 10–20% for elastic modulus changes of 2–3 fold, which appears inconsistent with our simulation predictions of negligible elastic contrast. However, their experimental setup involved external mechanical compression, which likely induced changes in optical properties (scattering coefficient, absorption coefficient) in addition to elastic properties. The observed signal changes may therefore reflect a combination of optical and mechanical effects, rather than pure elastic contrast.

Hai et al. (6; 7) demonstrated PA elastography in tissue phantoms and in vivo human subjects by applying focused ultrasound radiation force to induce localized tissue displacement. They reported measurable changes in PA signal amplitude correlated with tissue stiffness. Again, the interpretation of these results is complicated by potential confounding factors. Focused ultrasound can induce heating, cavitation, and nonlinear acoustic effects, all of which may alter optical and acoustic properties independently of elastic modulus.

Mandal et al. (13) used frequency-resolved PA imaging to assess elastic contrast in tissue-mimicking phantoms. They observed frequency-dependent signal variations that they attributed to elastic property differences. However, frequency-dependent effects can also arise from acoustic attenuation, dispersion, and scattering, which are not directly related to elastic modulus. Further theoretical and experimental work is needed to definitively separate elastic contrast from these confounding mechanisms.

4.4 Validation of k-Wave Simulation Framework

Our study demonstrates the utility of the k-Wave toolbox for comparative contrast mechanism studies in PA imaging. The pseudospectral time-domain methods implemented in k-Wave provide high accuracy and computational efficiency, enabling detailed investigation of signal generation and propagation physics (10; 12). The time reversal reconstruction capability allows direct assessment of image quality and reconstruction fidelity for different contrast scenarios.

The extension of k-Wave to elastic wave propagation via the `pstdElastic2D` function enables coupled simulation of acoustic and elastic wave dynamics. While our study focused on initial pressure generation rather than full elastic wave propagation, the framework can be extended to model more complex scenarios, such as:

- Shear wave generation by acoustic radiation force
- Mode conversion at tissue interfaces
- Coupling between compressional and shear waves in heterogeneous media
- Nonlinear elastic effects under large deformation

Such extensions would provide valuable insights into the interplay between optical, acoustic, and mechanical properties in PA imaging and related modalities.

4.5 Limitations and Future Directions

Several limitations of the current study should be acknowledged. First, we assumed a homogeneous background medium with uniform optical fluence. In realistic tissue scenarios, optical scattering and absorption lead to spatially varying fluence distributions, which can modulate the PA signal amplitude independently of elastic properties. Future simulations should incorporate coupled optical-acoustic modeling to account for these effects.

Second, we considered only two-dimensional geometries and linear array sensors. Three-dimensional simulations with more realistic sensor configurations (e.g., hemispherical arrays, tomographic geometries) would provide a more complete assessment of reconstruction fidelity and artifact characteristics. The k-Wave toolbox supports three-dimensional simulations via the `kspaceFirstOrder3D` function, enabling such extensions.

Third, we assumed lossless acoustic propagation. In biological tissues, acoustic attenuation and dispersion can significantly affect signal amplitude and frequency content, particularly at higher frequencies and longer propagation distances (12). Incorporating frequency-dependent attenuation would provide a more realistic model of in vivo PA imaging.

Fourth, our elastic contrast model assumed that variations in Young's modulus directly translate to variations in shear wave speed and Grüneisen parameter. In reality, the relationship between elastic modulus and PA signal generation may be more complex, involving stress-dependent changes in thermal expansion coefficient, nonlinear elastic effects, and viscoelastic relaxation. Experimental validation of the theoretical model is needed to confirm these assumptions.

Finally, we did not consider the effects of noise, limited dynamic range, and finite signal-to-noise ratio (SNR) in realistic PA imaging systems. The simulated elastic contrast signals are seven orders of magnitude weaker than absorption contrast signals, which would place extreme demands on detector sensitivity, amplifier dynamic range, and analog-to-digital converter resolution. A practical assessment of PA elastography feasibility should include detailed system modeling and noise analysis.

Future work should address these limitations through:

1. Coupled optical-acoustic simulations incorporating realistic tissue optical properties and fluence distributions
2. Three-dimensional simulations with clinically relevant sensor geometries
3. Incorporation of acoustic attenuation, dispersion, and nonlinear propagation effects
4. Experimental validation using tissue-mimicking phantoms with independently characterized optical and elastic properties
5. System-level modeling including noise sources, detector characteristics, and signal processing algorithms
6. Investigation of alternative elastic contrast mechanisms, such as stress-induced changes in absorption coefficient or thermal expansion coefficient

Despite these limitations, the current study provides valuable quantitative insights into the relative contributions of absorption and elastic contrast in PA imaging, and establishes a computational framework for further investigation of these mechanisms.

4.6 Practical Implications

The findings of this study have several practical implications for PA imaging and elastography:

1. PA imaging is fundamentally an absorption-based modality. The four-orders-of-magnitude difference in signal amplitude between absorption and elastic contrast confirms that PA imaging is primarily sensitive to optical absorption, not mechanical elasticity. This reinforces the established understanding of PA physics and validates the use of PA imaging for functional imaging applications (e.g., hemoglobin oxygenation, molecular contrast agents) that rely on absorption contrast.

2. Direct PA elastography is likely infeasible. The negligible amplitude of elastic contrast signals suggests that direct measurement of elastic properties via PA signal amplitude is impractical with current technology. Alternative approaches, such as PA-guided ultrasound elastography or multimodal imaging combining PA and elastography, may be more promising.

3. Confounding factors dominate elastic contrast. In experimental PA elastography studies, observed signal changes under mechanical loading are likely dominated by changes in optical properties, acoustic propagation paths, and motion artifacts, rather than genuine elastic contrast. Careful control experiments and theoretical modeling are essential to interpret such measurements.

4. k-Wave is a valuable tool for contrast mechanism studies. The k-Wave simulation framework provides a flexible and accurate platform for investigating PA signal generation and propagation physics. The ability to isolate individual contrast mechanisms and quantify their contributions is valuable for guiding experimental design and interpreting measurements.

5 Conclusion

This study presents a rigorous quantitative comparison of absorption-based and elasticity-based contrast mechanisms in photoacoustic imaging using the k-Wave pseudospectral simulation framework. We simulated two circular disc targets with identical geometry but differing physical properties: absorption coefficients of 1 cm^{-1} and 2 cm^{-1} (twofold variation), and Young's moduli of 94 kPa and 202 kPa (2.15-fold variation). Forward acoustic propagation was modeled using `kspaceFirstOrder2D`, elastic wave dynamics were computed via `pstdElastic2D`, and time reversal reconstruction was performed to recover source distributions.

The key findings are:

1. **Elastic contrast generates negligible PA signals.** Initial pressures for elastic contrast are approximately four orders of magnitude lower than those for absorption contrast, despite a larger relative variation in elastic modulus. This reflects the fundamental physics of PA signal generation, where the Grüneisen parameter for elastic contrast ($\sim 10^{-6}$) is vastly smaller than the bulk thermodynamic Grüneisen parameter ($\sim 10^{-1}$).
2. **Time reversal reconstruction faithfully recovers source positions.** Both absorption and elastic contrast scenarios exhibit accurate source localization and preservation of relative contrast ratios. However, reconstructed elastic pressures are seven orders of magnitude lower than reconstructed absorption pressures, rendering them effectively undetectable in practical imaging systems.
3. **PA imaging is fundamentally an absorption-based modality.** The dramatic difference in signal amplitude confirms that PA imaging is primarily sensitive to optical absorption, not mechanical elasticity. This validates the established understanding of PA physics and reinforces the use of PA imaging for functional and molecular imaging applications.
4. **Direct PA elastography faces significant challenges.** The negligible amplitude of elastic contrast signals suggests that direct measurement of elastic properties via PA signal amplitude is impractical with current technology. Experimental demonstrations of PA elastography likely reflect confounding factors (changes in optical properties, acoustic propagation paths, motion artifacts) rather than genuine elastic contrast.
5. **k-Wave provides a valuable platform for contrast mechanism studies.** The simulation framework enables isolation and quantification of individual contrast mechanisms, providing insights that guide experimental design and interpretation. The extension to elastic wave propagation via `pstdElastic2D` demonstrates the versatility of the k-Wave toolbox for multimodal imaging studies.

These findings clarify the relative contributions of optical and mechanical properties to PA signal generation and establish quantitative benchmarks for evaluating PA elastography approaches. While direct elastic contrast appears infeasible, alternative strategies such as PA-guided ultrasound elastography, multimodal imaging combining PA and elastography, or exploitation of stress-induced changes in optical properties may offer more promising avenues for integrating elastic property information into PA imaging.

Future work should address the limitations of the current study through coupled optical-acoustic simulations, three-dimensional geometries, incorporation of acoustic attenuation and dispersion, experimental validation with tissue-mimicking phantoms, and system-level modeling including noise and detector characteristics. Investigation of alternative elastic contrast mechanisms, such as stress-dependent changes in absorption coefficient or thermal expansion coefficient, may reveal conditions under which elastic properties contribute more significantly to PA signals.

In conclusion, this study demonstrates that photoacoustic imaging remains fundamentally an optical absorption-based modality, with elastic properties contributing negligibly to signal generation under physiologically relevant conditions. The k-Wave simulation framework provides a powerful tool for quantitative investigation of contrast mechanisms and validation of imaging system designs. These insights inform the develop-

ment of next-generation PA imaging technologies and guide the integration of complementary modalities for comprehensive tissue characterization.

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